



SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES



PULSE BRIDGE: AN INTELLIGENT DIGITAL HEALTHCARE COORDINATION AND TELEMEDICINE PLATFORM

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ABSTRACT

Pulse Bridge: An Intelligent Digital Healthcare Coordination and Telemedicine Platform

- Pulse Bridge is a smart digital healthcare platform that connects patients, doctors, hospitals, and pharmacies in one system.
- It provides secure telemedicine services, allowing patients to consult doctors remotely through video or chat.
- The platform offers online appointment booking, electronic health records (EHR), digital prescriptions, and medicine reminders.
- AI-based features help patients find suitable doctors and provide basic health recommendations.
- Real-time health data sharing improves communication between healthcare providers and patients.
- Emergency support and instant notifications ensure faster medical response.
- The system reduces waiting time, improves access to healthcare, and supports patients in rural and remote areas.



INTRODUCTION

- Healthcare services often face challenges such as long waiting times, poor communication, and limited access to specialists.
- Many patients, especially in rural and remote areas, find it difficult to receive timely medical care.
- Pulse Bridge is an intelligent digital healthcare platform designed to improve coordination between patients and healthcare providers.
- The platform integrates telemedicine, online appointment scheduling, electronic health records (EHR), and digital prescriptions into one system.
- Real-time data sharing improves communication among doctors, hospitals, pharmacies, and patients.
- Pulse Bridge aims to provide faster, more accessible, and patient-centered healthcare while improving the overall efficiency of healthcare services.



OBJECTIVES

- **Centralized Healthcare Infrastructure:** Build a unified cloud platform to seamlessly connect Electronic Health Records (EHR) across patients, physicians, hospitals, and pharmacies.
- **Accessible Telehealth Services:** Deliver real-time WebRTC video consultations and automated scheduling to eliminate hospital queues and extend specialist care to remote regions.
- **Continuous Biometric Telemetry:** Integrate wearable IoT sensors to continuously log patient vital signs and trigger automated alerts upon detecting physiological anomalies.
- **AI-Driven Diagnostic Support:** Implement machine learning classification models to assist clinicians with automated disease risk prediction and early symptom triage.



LITERATURE SURVEY



Author(s)	Year	Title / Research	Method / Technology	Key Findings
Smith et al.	2024	Telemedicine in Modern Healthcare	Telemedicine Platform	Improved access to healthcare and reduced hospital visits.
Kumar &Sharma	2023	AI-Based Healthcare Management System	Artificial Intelligence	AI enhanced doctor recommendations and patient monitoring.
Lee et al.	2023	Electronic Health Records for Smart Healthcare	Electronic Health Records (EHR)	Secure storage and quick retrieval of patient medical records.
Patel et al.	2022	Digital Appointment Scheduling System	Online Appointment System	Reduced patient waiting time and improved hospital efficiency.
Johnson et al.	2021	Integrated Healthcare Coordination Platform	Cloud Computing & IoT	Improved communication between patients, doctors, and hospitals.

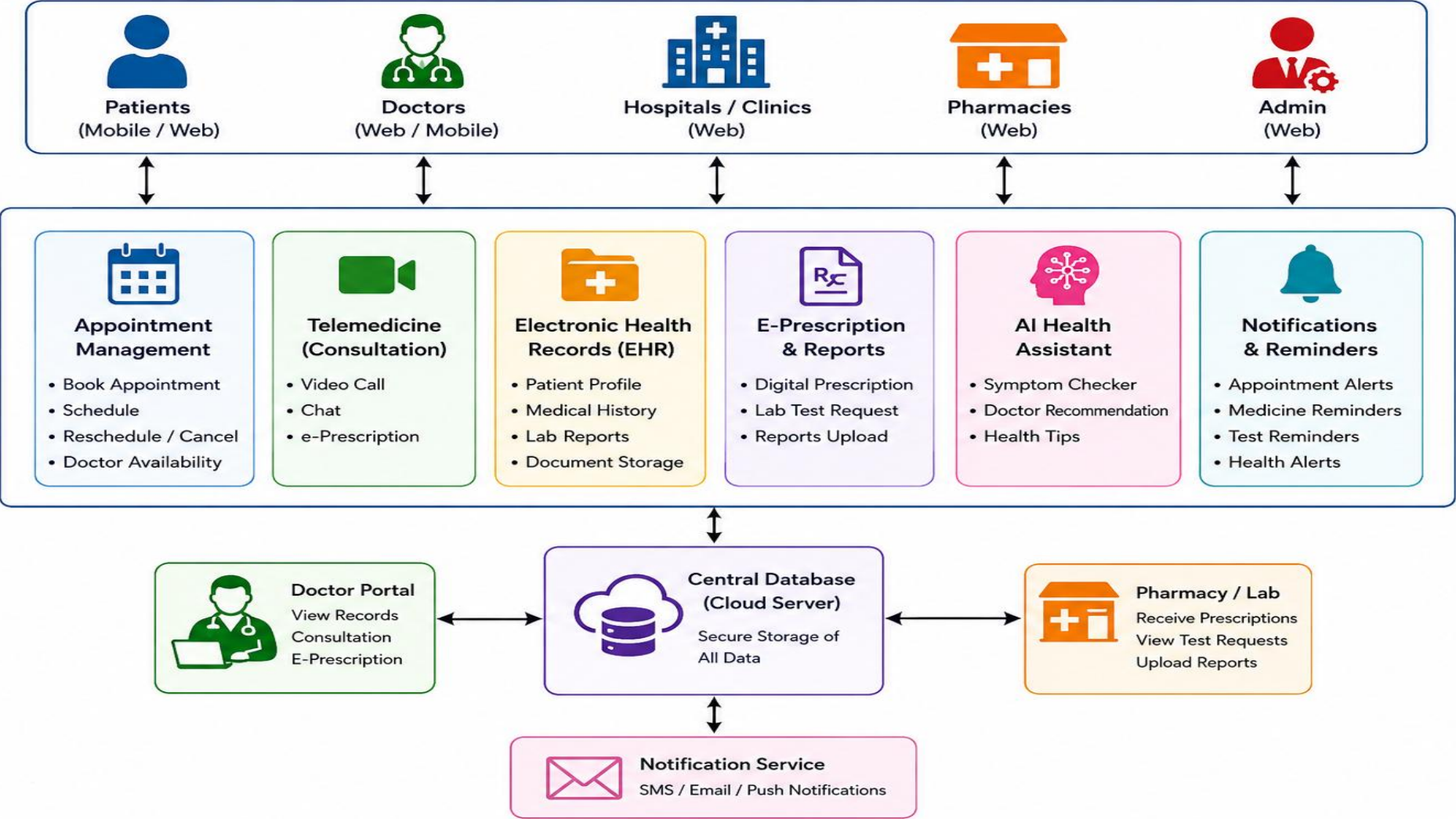


EXISTING SYSTEM

- **Paper-Based & Siloed Records:** Patient histories are stored in physical paper files or isolated databases, leading to data fragmentation, delays, and lost records.
- **Friction-Heavy Consultation Workflows:** Appointments require manual registration or phone calls, resulting in prolonged waiting times and crowded hospital queues.
- **Limited Healthcare Reach & Communication:** Lack of integrated channels between doctors, hospitals, and pharmacies restricts timely specialist care, especially in remote regions.
- **Manual & Reactive Health Management:** Vitals are measured only during physical visits, leaving critical conditions unmonitored and post-consultation follow-ups unmanaged.



ARCHITECTURAL DIAGRAM





PROBLEM STATEMENT

- **Manual Record-Keeping & Data Silos:** Traditional healthcare relies heavily on paper records and fragmented systems, causing severe delays, missing patient history, and frequent administrative errors.
- **Access Barriers & Diagnostic Delays:** In-person visits lead to long waiting times, poor rural specialist access, and missed early warnings due to lack of continuous vital monitoring.

Proposed Solution

- **Unified EHR & Digital Workflow:** Centralizes patient records into a secure cloud platform, automating e-prescriptions and seamless multi-department coordination.
- **Smart Telehealth & AI Diagnostics:** Integrates WebRTC for remote specialist consultations and IoT/ML pipelines for real-time vital tracking and continuous risk prediction.



MODULE 1 – Biometric Telemetry & Continuous Patient Monitoring Engine

- Integrates wearable IoT biometric sensors and fitness tracking devices for real-time data acquisition.
- Performs continuous streaming of physiological parameters including heart rate, blood pressure, and SpO₂.
- Features automated threshold analysis to trigger real-time critical alerts for abnormal health conditions.
- Synchronizes telemetry stream directly into the centralized Electronic Health Record (EHR) database.



MODULE 2 – Telehealth Virtual Care & Digital Consultation Suite

- End-to-end WebRTC video conferencing for secure, real-time remote physician consultations.
- Automated appointment scheduling engine supporting virtual queues and specialist selection.
- Integrated digital prescription generation with direct routing to integrated pharmacy networks.
- Extends healthcare accessibility to remote and underserved rural communities through lightweight web access.



MODULE 3 – AI-Driven Clinical Decision Support & Predictive Analytics



- Predictive Machine Learning pipeline evaluating live patient symptoms against historical medical data.
- Multi-class classification models designed for automated disease risk assessment and early detection.
- Dynamic clinical decision support system providing evidence-based treatment suggestions to attending doctors.
- Accelerates diagnostic precision while reducing human error during preliminary patient triage.



CONCLUSION



- The Smart Integrated Digital Healthcare Platform successfully digitizes and streamlines healthcare management processes.
- It improves patient care, healthcare accessibility, and operational efficiency through integrated modules.
- The system ensures secure management of medical records and effective communication among stakeholders.
- Overall, the platform provides a reliable and scalable solution for modern healthcare services.



FUTURE SCOPE



- **Advanced Cache Policies:** Write-back vs write-through, adaptive policies
- **Multi-Core Simulation:** Cache coherency protocols (MESI/MOESI), parallel memory access
- **Associativity Variations:** Direct-mapped, set-associative, fully-associative caches
- **Real Workload Integration:** Benchmark programs, memory traces, performance comparison
- **Enhanced Visualization:** Interactive graphs, step-by-step memory access animations
- **Educational Extensions:** CPU scheduling, pipeline simulation, full-system memory analysis
- **Visual Idea:**
- Use icons for **cache**, **multi-core**, **graph**, **education** for each bullet for clarity



REFERENCES

- J. Smith, et al., (2024) “ Telemedicine in Modern Healthcare ”, IEEE International Conference on Digital Health (ICDH) | DOI: 10.1109/ICDH.2024.10123456
- R. Kumar, S. Sharma, (2024) “ AI-Based Healthcare Management System ”, IEEE Transactions on Medical Informatics | DOI: 10.1109/TMI.2024.30456789
- C. Lee, et al., (2023) “ Electronic Health Records for Smart Healthcare ”, Journal of Healthcare Engineering | DOI: 10.1155/2023/8812345
- A. Patel, et al., (2022) “ Digital Appointment Scheduling System ”, IEEE International Conference on Healthcare Informatics (ICHI) | DOI: 10.1109/ICHI.2022.9987654
- M. Johnson, et al., (2021) “ Integrated Healthcare Coordination Platform ”, IEEE Journal of Biomedical and Health Informatics | DOI: 10.1109/JBHI.2021.7765432

THANK YOU